

```

1 // Define (p0, v0, t0), ..., (p4, v4, t4) as in the previous figure.
2 StateData state0(p0, v0, t0); // Defined similarly for all other states.
3 // Assume all the poses are in SE(3).
4
5 // Suppose we have a binary factor between p1 and p3.
6 // Assume relPose_13 is the relative pose measurement between p1 and p3
7 const auto binaryNoiseModel =
8     noiseModel::Diagonal::Sigmas(Vector6::Ones());
9 const auto binaryFactor = std::make_shared<BetweenFactor<Pose3>>(
10     p1, p3, relPose_13, binaryNoiseModel);
11
12 // Suppose we also have a prior (unary factor) on p1.
13 const auto p1PriorValue = Pose3(); // origin
14 const auto unaryNoiseModel =
15     noiseModel::Diagonal::Sigmas(Vector6::Ones());
16 const auto unaryFactor = std::make_shared<PriorFactor<Pose3>>(
17     p1, p1PriorValue, unaryNoiseModel);
18
19 // Suppose all poses are connected via WNOA factors.
20 auto Q_se3 = Vector6::Ones(); // Power spectral density
21
22 // Now, instead of defining WNOA factors and adding all the states into
23 // the graph, we want state1 and state3 to be interpolated.
24
25 // Factor graph visualization before interpolation:
26 //      unary
27 //      |
28 //  0 ---- 1 ---- 2 ----- 3 ----- 4
29 //  b      i      b      i      b
30 //      --- between ---
31
32 // Define the set of bordering states that will remain in the main solve.
33 set<StateData> borderStates = {state0, state2, state4};
34
35 // All other states will be interpolated.
36 set<StateData> interpStates = {state1, state3};
37
38 // Define the interpolation factors to wrap the two original factors.
39 // This also adds the WNOA factors to the states.
40 const auto binaryFactorWrapped = WnoaInterpFactor<Pose3>(
41     binaryFactor, borderStates, interpStates, Q_se3);
42 const auto unaryFactorWrapped = WnoaInterpFactor<Pose3>(
43     unaryFactor, borderStates, interpStates, Q_se3);
44
45 // Create graph.
46 NonlinearFactorGraph graphInterp;
47 graphInterp.add(binaryFactorWrapped);
48 graphInterp.add(unaryFactorWrapped);

```